

WEEK – 4

Divyadharshini G

Project:

stock Market Analysis using Power BI

1. Simple Moving Average Using DAX

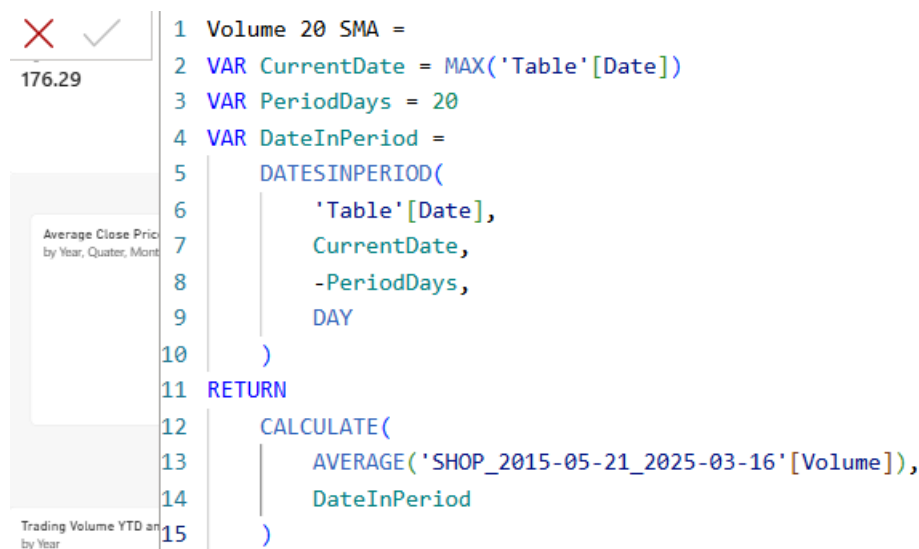
Two moving-average measures were created using DAX:

- 20-Day SMA **Measure Name:** Volume 20 SMA

It calculates the average trading volume over the previous 20 days.

- 50-Day SMA **Measure Name:** Volume 50 SMA

It calculates the average trading volume over the previous 50 days.



```

1 Volume 50 SMA =
2 VAR CurrentDate = MAX('Table'[Date])
3 VAR PeriodDays = 50
4 VAR DateInPeriod =
5     DATESINPERIOD(
6         'Table'[Date],
7         CurrentDate,
8         -PeriodDays,
9         DAY
10    )
11 RETURN
12     CALCULATE(
13         AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume]),
14         DateInPeriod
15    )

```

2. Moving Average Crossover Signal

A DAX measure named MA Crossover Signal was created.

It compares the 20-day SMA with the 50-day SMA.

- When 20 SMA > 50 SMA, the signal is displayed as Bullish (20 > 50).
- Otherwise, it is displayed as Bearish (20 < 50).

```

1 MA Crossover Signal =
2 IF(
3     ISBLANK([Volume 20 SMA]) ||
4     ISBLANK([Volume 50 SMA]),
5     BLANK(),
6     IF(
7         [Volume 20 SMA] > [Volume 50 SMA],
8         "Bullish (20 > 50)",
9         "Bearish (20 < 50)"
10    )
11 )

```

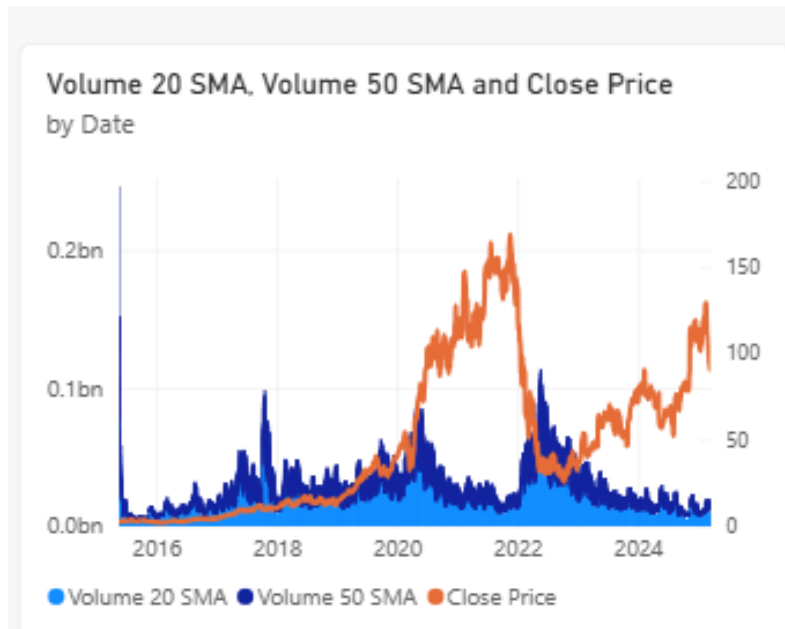
3. Price Trend Visualization

A Line and Stacked Column Chart was created to visualize stock price and moving-average trends.

The chart contains:

- Date on the X-axis

- Close Price
- 20 SMA
- 50 SMA

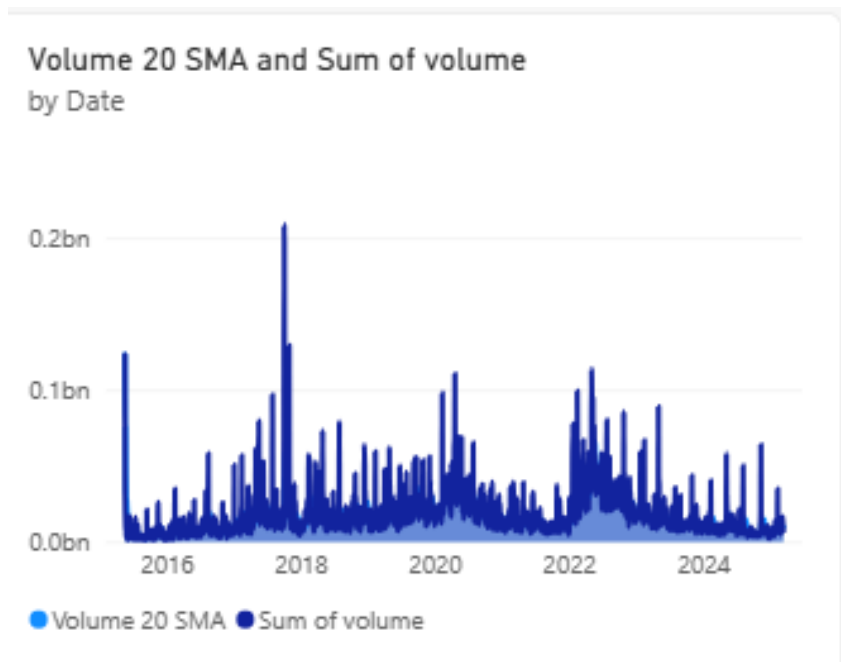


4. Trading Volume Analysis

A Clustered Column Chart was created to analyze trading volume.

The chart contains:

- Date on the X-axis
- Trading Volume / Total Volume
- Volume 20 SMA



5. Conditional Formatting for Volume

A Volume Color DAX measure was created to highlight significant volume changes.

Conditional formatting was then applied to the volume columns using the FX option.



Color - Categories

Format style

Field value



What field should we base this on?

Volume Color

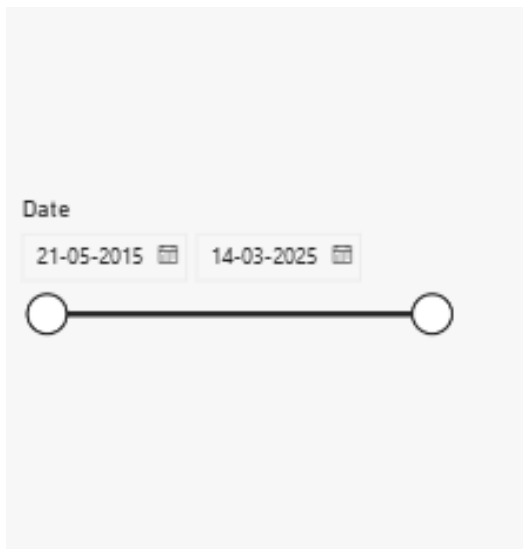


6. Date Range Slicer

A Date Range Slicer was created using the Date field.

The slicer was configured in **Between** mode so that a specific date range can be selected.

This allows the user to interactively filter the stock analysis for a selected period.



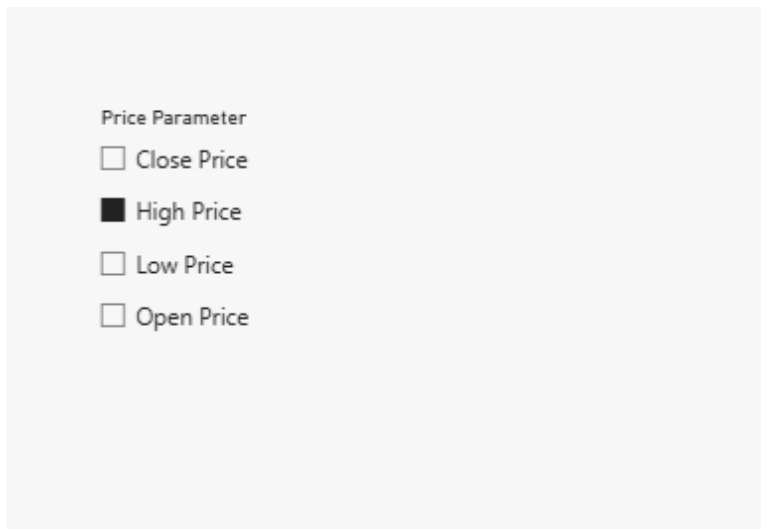
7. Field Parameter – Price Selector

A Field Parameter was created using:

- Close Price
- High Price
- Low Price
- Open Price

The parameter was added as a slicer. When a different price is selected, the price line in the main chart changes accordingly.

This makes the report interactive and allows different price measures to be analyzed without creating separate charts.



8. Custom Tooltip Page

A separate report page was created.

The page was configured with:

- **Allow use as tooltip → On**
- **Canvas Type → Tooltip**

The tooltip contains stock information such as:

- Open Price
- High Price
- Low Price
- Close Price
- Trading Volume
- Daily Price Change

Parameters

×

Add parameters to visuals and DAX expressions so people can use slicers to adjust the inputs and see different outcomes. [Learn more](#)

What will your variable adjust?

Fields

Name

Price Parameter

Add and reorder fields

Close Price

High Price

Low Price

Open Price

☒ Add slicer to this page

Fields

Search

☐ Average Close Price

☒ Close Price

☐ Close Price LY

☐ Close Price PY

☐ Close Price YTD

☐ Daily Price Change

☐ Daily Return %

☒ High Price

☒ Low Price

☐ MA Crossover Signal

☒ Open Price

☐ Previous Day Close

Create

Cancel

9. Final Power BI Dashboard

The completed Week 4 report contains interactive visualizations for:

- Stock price trends
- Moving averages
- Trading volume
- Volume moving average
- Volume spike highlighting
- Date filtering
- Dynamic price selection

- Custom stock information tooltips

